

SAiDL Assignment 2026

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Core-ML Task

1 Intro

This is my report on the Core-ML task of Attention-based Transformer, trained on WikiText-2. I'll be updating my progress here.

2 Basic Transformer Baseline

From the 'Attention is All You Need' article, the core of the Transformer is the Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where:

- $Q \rightarrow$ Queries
- $K \rightarrow$ Keys
- $V \rightarrow$ Values
- $d_k \rightarrow$ Dimension of the keys

3 Baseline Model Transformer Details

I've used AI to explain what each class does in the Transformer code.

3.1 Data Processing and Embeddings

InputEmbedding: Converts input tokens into dense vectors of dimension d_{model} and scales them by $\sqrt{d_{model}}$ to stabilize gradients. It maps the discrete vocabulary to a continuous space suitable for neural network processing.

PositionalEncoding: Injects information about the relative or absolute position of tokens using sine and cosine functions of different frequencies. This allows the model to utilize the order of the sequence, as the self-attention mechanism itself is permutation-invariant.

3.2 Model Components

LayerNormalization: Standardizes the inputs across the features dimension to a mean of zero and variance of one, followed by a learnable affine transformation. This ensures stable training and faster convergence by mitigating internal covariate shift.

FeedForwardBlock: Consists of two linear transformations with a ReLU activation and dropout in between to process each position independently. It increases the model's capacity to learn complex patterns within the hidden representations.

MultiHeadAttentionBlock: Projects the Q , K , and V into multiple subspaces to attend to information from different representation aspects simultaneously. It uses the previously-mentioned scaled dot-product attention in parallel across h heads before concatenating and re-projecting the results.

3.3 Parameters Used for WikiText-2 Training

The baseline Transformer was trained on the WikiText-2 dataset using the following configuration:

Table 1: Hyperparameters for Baseline Transformer Training

Parameter	Value	Description
<i>Model Architecture</i>		
d_model	128	Embedding dimension
n_heads	4	Number of attention heads
n_layers	4	Number of decoder layers
d_ff	512	Feed-forward hidden dimension
dropout	0.1	Dropout rate
<i>Sequence</i>		
seq_len	1024	Context window length (tokens)
vocab_size	50,257	GPT-2 BPE tokenizer vocabulary size
<i>Training</i>		
batch_size	8	Sequences per batch
grad_accum_steps	4	Gradient accumulation steps
effective_batch	32	Effective batch size (batch $\times$ accum)
lr	3×10^{-4}	Peak learning rate (AdamW)
weight_decay	0.1	AdamW weight decay
warmup_steps	200	Linear warmup steps
max_steps	5,000	Total training steps
eval_interval	250	Steps between validation evaluations
eval_iters	50	Batches averaged per validation estimate

3.4 Results

Table 2 summarizes the training and validation metrics recorded every 250 steps over the full 5,000-step training run on WikiText-2.

The model converged to a best validation perplexity of **224.70** at step 5,000, with a corresponding training perplexity of 95.02. The gap between training and validation perplexity indicates mild overfitting, which is expected given the relatively small model size (13.71M parameters) and limited dataset. Throughput remained highly stable throughout training at approximately **29,000 tokens/sec** on an NVIDIA Tesla T4 GPU, demonstrating consistent hardware utilization. Validation perplexity improved rapidly in the early stages (dropping from 1759 to 255 in the first 2,500 steps) before plateauing around 229–232 for the final 1,500 steps, suggesting the model had largely converged by step 3,500.

Table 2: Training and Validation Metrics Over 5,000 Steps

Step	Train Loss	Train PPL	Val Loss	Val PPL	Tok/s
250	7.44	1696.88	7.47	1759.35	29,580
500	6.60	735.60	6.71	817.41	29,258
750	6.08	436.26	6.23	508.59	29,118
1000	5.76	316.06	6.03	417.27	29,117
1250	5.53	251.85	5.87	353.61	29,033
1500	5.33	206.69	5.77	319.71	29,094
1750	5.15	172.62	5.66	288.23	29,044
2000	5.03	153.57	5.62	276.00	29,040
2250	4.92	137.67	5.57	261.15	29,130
2500	4.86	129.11	5.54	255.69	29,052
2750	4.73	112.97	5.48	239.11	29,046
3000	4.70	109.93	5.48	240.87	29,107
3250	4.64	103.94	5.45	232.15	29,115
3500	4.61	100.23	5.44	231.49	29,081
3750	4.56	95.45	5.45	232.93	29,015
4000	4.56	96.05	5.44	230.93	29,011
4250	4.54	94.15	5.43	229.01	29,072
4500	4.56	95.62	5.45	232.38	29,101
4750	4.51	90.87	5.45	231.87	29,111
5000	4.55	95.02	5.41	224.70	29,034